

COMP544 – Machine Learning – Project

Objective. In this project you will have to:

- Design
- Implement
- Evaluate and Optimize

a machine learning model to be applied for **Emotion classification**.

Learning Outcomes. Through this project, the student will get familiar with experimental evaluation of machine learning models, pre-processing of data, writing a short technical report, selecting parameters of a model, combining classifiers, use of appropriate libraries (Python), and dealing with a real-world machine learning problem.

Teams. This project should be done in teams of **2 or 3 students** (not 1, not 4, etc).

Data. You can find the data uploaded to moodle. The file includes a set of documents marked with the following labels: *sadness, neutral, worry, happiness, fun, surprise, hate*. The data are in raw format (text) so you have to convert them to the form that the algorithms can process. This is part of the pre-processing step.

Goal. You have to train, optimize and evaluate the model on these data in order to get the best possible predictive performance in new, unknown data.

Experimenting and Selecting your model. You are provided with a file named `train.csv`. Use this to select the best model for this type of data in your opinion. For that, you need to experiment with multiple algorithms (that we have seen in class) and parameters (as we have seen in class) for each algorithm and with proper evaluation process (train-test, cross-validation or whatever you think is best). This experimentation should be presented in your Technical Report (see below for details).

Preparing your model for evaluation. After you decide on the best model, you need to prepare it to classify some new, unknown data. The new data will not be provided to you. The instructors should be able to use your model in order to classify new data. The instructors will have a `test.csv` file which will have an identical format as the `train.csv` file, but with no labels (that column will be missing). Your model (process) should load that data, classify and store the decision in a `predictions.txt` file. One prediction per line.

The `predictions.txt` file should look like this:

```
sadness
neutral
neutral
worry
sadness
fun
. . .
```

Obviously this file will have as many lines as the `test.csv` file.

Evaluation (by the instructors)

The instructors having:

- a) The `predictions.txt` of your model.
- b) The actual (correct) labels of the new data (that you don't have).

Will calculate the **accuracy** of your approach.

Based on the accuracy, your approach will be ranked in comparison to the other teams in the course. This ranking will affect one part of your project grade (see below).

Deliverables

For this project, you will have to submit on moodle (only one member of the team):

1. Your trained model.
 - a. Python - you should provide the files with all your code. One of your files should be named `main.py`. In this file, you should have a function that will be named `train_test(train.csv, test.csv)` taking as parameters the `train.csv` file (that is provided to you) and the `test.csv` file (that is not provided to you), so that the evaluators can easily run it and get results (i.e. the `predictions.txt` file).

Python: If you think it is a good idea to add a file with instructions on how to run your process/code to the instructors please add a `readme.txt` file to your submission.

2. A technical report

Describe in 4 pages how you have decided to use the model you have selected. What experiments you did, which models you tried out, what you observed, how you did the evaluation, why you selected that particular setting in the end and whatever else you think necessary. You are advised to professionally present your results with tables and plots.

Marks - Overview: Total: 100 points, out of which:

- A. 15 points: pre-processing
- B. 50 points: model evaluation (experiments, tuning, argumentation, links with theory, etc.)
- C. 20 points: quality of technical report (presentation of results, plots, tables, etc.)
- D. 15 points: your rank

Marks - Details

- **About point B:** You must write a Technical Report describing your method and explaining why you decided to select that particular model, with these specific parameters. Length: 4 pages (maximum).
- **About point D:** Your points will obtained based on the following equation:
$$D = 15 - (\text{Rank} - 1) * 2$$
So if you rank 1st you get the full 15 points. If you rank 2nd you get 15-2=13 points, if you rank 8th, you get 2 points. (This formula might slightly change depending on the number of the teams)

Interviews: If necessary, the instructor might ask you for a short meeting to discuss and explain your approach and understand the contributions' of each team member. This measure might be applied to a small group of teams only.